



Faculty of Engineering and Technology
Department of Electrical and Computer Engineering

Linux Labortary
ENCS3130
Project # 2
----- Test Cases -----

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Teaching Assistant: Eng. Loor Sawalhi
Section: 4
Date: 18/1/2026

Input files:

1. Schedules

```
120345, Ahmad, Saleh
S | [9-10]ENCS334; [10-11]ENCS324; [13-15]OH;
M | [9-10]ENCS334; [10-11]ENCS324; [13-15]OH;
T | [8-11]ENCS212; [11-12]OH; [13-15]ENCS333;
W | [9-10]ENCS334; [10-11]ENCS324; [13-15]OH;
Th | [11-12]OH; [13-15]ENCS333;

1221669, Wadee, Naser
S | [8-11]ENCS314; [10-12]ENCS53; [13-14]OH
M |
T |
W | [9-10]ENCS535
Th | [8-11]ENCS314; [11-12]ENCS534; [12-13]ENCS532; [10-12]OH
```

2. Valid corse codes

1	ENCS334
2	ENCS324
3	ENCS212
4	ENCS333
5	ENCS314
6	ENCS311

3. Instructors preferences

120345:	ENCS334; ENCS324; ENCS212; ENCS333; ENCS312; ENCS311
1221669:	ENCS334; ENCS533; ENCS233; ENCS43

Output result:

```
Validating Dr. Ahmad Saleh Schedule

1. Validating Teaching Load .....
Valid Teaching Load Of 13.0 Hours

2. Validating Office Hours Ratio .....
Valid Office Hours Ratio For That Specific Load

3. Validating Teaching Days Count .....
Valid Teaching Days Distribution

4. Validating If Time Conflict Exists .....
0 Overlaps Detected!

5. Validating Consecutive Teaching Rule .....
No More Than Two Courses/Labs Are Scheduled Consecutively Without A Break.

6. Validating Allowed Teaching Days .....
Courses Are In Valid Days

7. Course And Lab Code Validation .....
All Courses Codes Are Valid

8. Instructor Preference Validation .....
All Courses Codes Are Preferred By This Instructor
```

Validating Dr. Wadee Naser Schedule

1. Validating Teaching Load

Invalid Teaching Load. It Is 11.0 Which Is Less Than 12 Hours

2. Validating Office Hours Ratio

Invalid Office Hours Ratio: 27%

3. Validating Teaching Days Count

Invalid Teaching Days Distribution

4. Validating If Time Conflict Exists

Found An Overlap Between [08:00 - 11:00] ENCS314 And [10:00 - 12:00] ENCS53 in Day Symbol Saturday

Found An Overlap Between [12:00 - 13:00] ENCS532 And [10:00 - 12:00] OH in Day Symbol Thursday

2 Overlaps Detected!

5. Validating Consecutive Teaching Rule

More than 2 consecutive courses are detected in Thursday

6. Validating Allowed Teaching Days

Invalid; Involved Courses: Encs314

7. Course And Lab Code Validation

The Following Codes Are Not Valid

ENCS53

ENCS532

ENCS535

ENCS534

8. Instructor Preference Validation

The Following Codes Are Not Preferred By This Instructor

ENCS53

ENCS534

ENCS532

ENCS535

ENCS314